

Mate 6551

Tarea 3

- *3.1. Generalize Theorem 1.3 as follows. Let $A \subset X$ and $B \subset Y$ be given. Assume that $f_0, f_1: X \rightarrow Y$ with $f_0|_A = f_1|_A$ and $f_i(A) \subset B$ for $i = 0, 1$; assume $g_0, g_1: Y \rightarrow Z$ with $g_0|_B = g_1|_B$. If $f_0 \simeq f_1 \text{ rel } A$ and $g_0 \simeq g_1 \text{ rel } B$, then $g_0 \circ f_0 \simeq g_1 \circ f_1 \text{ rel } A$.
- *3.5. Let $f_0 \simeq f_1 \text{ rel } \mathbf{I}$ and $g_0 \simeq g_1 \text{ rel } \mathbf{I}$ be paths in X and Y , respectively. If, for $i = 0, 1$, (f_i, g_i) is the path in $X \times Y$ defined by $t \mapsto (f_i(t), g_i(t))$, prove that $(f_0, g_0) \simeq (f_1, g_1) \text{ rel } \mathbf{I}$.
- 3.12. If $\pi_1(Y, y_0) \neq \{1\}$ for some pointed space (Y, y_0) , then $\pi_1(S^1, 1) \neq \{1\}$. (*Hint:* Otherwise 1_S is nullhomotopic, where 1_S is the identity map on S^1 , and this implies that $f = f \circ 1_S$ is nullhomotopic for every closed path f in Y at y_0 .)
- 3.17. Prove that S^1 is not a retract of D^2 .
- 3.19. Let f be a closed path in S^1 at 1.
- (i) If f is not surjective, then $\deg f = 0$.
 - (ii) Give an example of a surjective f with $\deg f = 0$.